

30-day AI Engineer interview preparation roadmap ▾

Length*: 39,469 tokens | Cost: \$0.29 | Cached for: 27m

Create a 30-day learning roadmap.

Context:

- Current Situation: M.Tech professional with ~2 years experience, actively giving AI Engineer interviews.
 - Current Skills: LLMs, ML, Python, SQL, AI, RAG, LangChain, LangGraph, FastAPI, Prompt Engineering, Statistics etc.
 - Goal: Secure an AI Engineer job in a month with my desired skill set and good pay.
 - Available Time: 4-5 hours per day.
 - Experience Level: Intermediate.
 - Preferred Learning Style: Videos, Reading material and projects.
- Include:
- Weekly milestones
 - Daily tasks
 - Resources
 - Projects
 - Final outcome Make it practical and tailored to my experience level.
- Make roadmap in one page in the form of an image so it can be shared as an image in a linkedin post. make it professional and use eye catchy template

Show less

Reading frontend design skill for creating a professional visual roadmap >

30-Day AI Engineer Roadmap

M.TECH → AI ENGINEER · 4-5 HRS/DAY · INTERMEDIATE LEVEL

Secure your target role in 30 days — structured, practical, project-driven

WEEK 1

Week 1 (Days 1–7) — Foundation Audit & Interview Positioning

Milestone: Portfolio live · GitHub refreshed · 3 target JDs mapped

Day 1 **Audit & Gap Analysis**

Review 20 JDs · list missing skills · score yourself on LLM, RAG, MLOps, System Design

Day 2 **LLM Internals Deep Dive**

Watch: Andrej Karpathy "Let's build GPT" · Read: Attention is All You Need (annotated)

Day 3 **RAG Architecture Mastery**

Chunking strategies, vector DBs (Qdrant/Pinecone), reranking, hybrid search — revise & code

Day 4 **LangChain / LangGraph Advanced**

Agents, tool calling, memory, multi-agent workflows · LangGraph Docs + YouTube (Greg Kamradt)

Day 5 **FastAPI + Deployment Patterns**

Async endpoints, background tasks, Docker, HuggingFace Spaces deploy · Build a mini AI API

Day 6 **GitHub Portfolio Sprint**

Clean repos, add READMEs with architecture diagrams, pin 4–5 best AI projects

Day 7 **Project 1 — Production RAG Chatbot**

PDF ingestion → chunking → vector DB → LangChain QA chain → FastAPI → de

 PROJECT 1
RAG Chatbot

WEEK 2

Week 2 (Days 8–14) — Advanced AI Systems & MLOps

Milestone: MLOps pipeline live · Fine-tuning done · System Design patterns practiced

Day 8 **Fine-Tuning & PEFT**

LoRA, QLoRA, RLHF basics · HuggingFace PEFT library · Fine-tune LLaMA on Colab

Day 9 **Evaluation & Observability**

RAGAS, LangSmith, TruLens, LLM-as-judge · Build eval pipeline for your RAG project

Day 10 **MLOps & CI/CD for AI**

MLflow, DVC, GitHub Actions pipeline · Read: Made With ML (Goku Mohandas)

Day 11 **Vector DBs & Embeddings Deep Dive**

FAISS, Chroma, Weaviate, Pinecone · Embedding models (BGE, E5, OpenAI) · Benchmarks

Day 12 **AI System Design Patterns**

Design: search engines, recommendation, chatbots, classification · ByteByteGo AI chapter

Day 13 **Cloud AI Services (AWS/GCP/Azure)**

Bedrock, Vertex AI, OpenAI API · Cost optimization, rate limits, fallback strategies

Day 14 **Project 2 — Multi-Agent AI System**

LangGraph agents: Researcher + Writer + Critic · Human-in-loop · LangSmith tra

 PROJECT 2
Multi-Agent System

WEEK 3

Week 3 (Days 15–21) — Interview Prep & Mock Drills

Milestone: 50+ interview questions answered · Coding streak active · Elevator pitch ready

Day 15 **ML & Statistics Interview Questions**

Bias-variance, regularization, Bayes, A/B testing, p-values · ML Zoomcamp notes

Day 16 **LLM-Specific Interview Questions**

Hallucination, context window, tokenization, RLHF, constitutional AI · Chip Huyen's LLMops

Day 17 **Coding Round — LeetCode AI Focus**

10 medium problems: arrays, strings, DP · Python data structures for ML interviews

Day 18 **System Design — AI Scale**

Design: ChatGPT-like API, real-time recommendation, feature store · Latency/cost tradeoffs

Day 19 **Mock Interview — Behavioural + Technical**

STAR stories for AI projects · Use Pramp / interviewing.io · Record yourself

Day 20 **Resume & LinkedIn Optimization**

Keyword-optimize for AI Engineer roles · Add impact metrics · Request 3 recommendations

Day 21 🚀 **Project 3 — End-to-End LLM App**

Finetune + RAGAS eval + MLflow tracking + FastAPI + Docker + CI/CD (GitHub)

PROJECT 3
E2E LLM App

WEEK 4

Week 4 (Days 22–30) — Apply, Negotiate & Land the Offer

Milestone: 30+ applications sent · 5+ interviews booked · Offer in hand

Days 22–23 **Targeted Applications Blitz**

LinkedIn · Naukri · Instahyre · Wellfound · Direct referrals · 15 apps/day min

Days 24–25 **Live Interview Rounds**

Take all available interviews · Note patterns · Re-read your projects before each call

Days 26–27 **Take-Home / Case Study Prep**

Template repo ready · Time-box 4 hrs per challenge · Write a brief design doc with trade-offs

Day 28 **Offer Evaluation & Salary Negotiation**

Benchmark: Levels.fyi, Glassdoor · Counter-offer script · Equity vs cash analysis

Days 29–30 🎯 **Final Push — LinkedIn Content + Networking**

Post project writeups · Engage AI Engineer communities · Follow up on all pending loops

KEY RESOURCES

Karpathy YouTube

Chip Huyen — LLMOps

LangChain Docs

Made With ML

RAGAS + LangSmith

ByteByteGo AI

Pramp / Interviewing.io

HuggingFace PEFT

🎯 **Final Outcome**

3 shipped projects · Polished portfolio · 30+ applications · AI Engineer offer @ target salary

Consistency beats intensity · 4–5 hrs/day · Ship, document, apply · You've got this.